# AKASH JADON

## LEAD AI SYSTEMS ARCHITECT & FOUNDING FULL-STACK ENGINEER

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## PROFESSIONAL SUMMARY

Senior AI Systems Architect and High-Agency Full-Stack Engineer with 3+ years engineering autonomous LLM orchestration kernels, distributed concurrency pipelines, and production SaaS architectures. Architect of Astra AI Core (4,400+ LOC custom Python streaming runtime with per-provider FSM circuit breakers and live token interceptors), BossConsole (native JVM/Kotlin desktop harness executing 100+ MCP tools with sub-110MB RAM footprint), and DevisBat (commercial Swiss trade fintech platform with Gemini Vision estimation and QR-bill generation). Deep expertise in low-latency token streaming (sub-120ms TTFT), atomic Redis locking (SETNX), and production-hardened web applications on Next.js 15, TypeScript, and AWS.

## TECHNICAL ARSENAL

| | |
|---|---|
| **Agentic AI & Orchestration:** | Autonomous ReAct execution loops, Model Context Protocol (MCP), streaming <think> token state machines, per-provider FSM circuit breakers, RAG, Swappable LLMs (Claude 3.5 Sonnet, Gemini 2.5, DeepSeek-V3, Groq). |
| **Concurrency & Reliability:** | Distributed Redis atomic locks (SETNX), SHA-256 canonical idempotency hashing, token-bucket rate limiting, exponential backoff retries with jitter, Dead-Letter Queues (DLQ), zero-hallucination pipelines. |
| **Full-Stack & Cloud Infra:** | Next.js 15 (App Router, Server Actions), React 19, TypeScript, Tailwind CSS, Python (FastAPI, asyncio, Uvicorn), PostgreSQL (Row-Level Security), Supabase, Redis, Docker, AWS (ap-south-1), Vercel Edge Runtime. |
| **Security & Protocols:** | AES-256-GCM record-level encryption, Webhook HMAC validation, Meta WhatsApp Cloud API, Telegram Business API, Meta Conversions API (CAPI), Close CRM API, Capacitor Android SDK native compilation. |

## REPRESENTATIVE PRODUCTION ARCHITECTURES

### Astra AI Core — Autonomous AI Runtime & Streaming Kernel

**Python / FastAPI / Playwright / SSE / Supabase**

- Kernel Architecture & Sub-120ms TTFT: Engineered a 4,400+ LOC custom Python streaming backend utilizing Server-Sent Events (SSE) and async connection pooling, achieving sub-120ms Time-To-First-Token across 4 model providers.
- FSM Circuit Breaker Failover: Built Finite State Machine circuit breakers monitoring real-time error rates across NVIDIA GLM, DeepSeek, Gemini, and Groq; slashed streaming dropout rates from 14.2% to <0.05% with zero user interruption.
- Streaming <think> Token Interceptor: Built real-time token state machine intercepting reasoning tags on live streams, stripping client clutter, rendering collapsible UI accordions, and triggering automated Playwright web research upon uncertainty detection.
- Autonomous Sandboxed ReAct: Deployed self-healing browser automation loops inside headless Playwright sandboxes backed by dual-key rotation vaults and 500-slot semantic LRU response caching.

### BossConsole — Native JVM Multi-Threaded AI Agent Operator Console

**Kotlin / Compose Desktop / JVM 21 / MCP / Open Source**

- Native Desktop Runtime (110MB RAM): Architected an open-source JVM/Kotlin Compose Desktop agent operator console, cutting resident memory from 850MB+ (typical Electron wrappers) down to <110MB while orchestrating Claude Code, Gemini, and Codex.
- 100+ MCP Tool Harness: Implemented hot-reloadable Model Context Protocol (MCP) server/client harness enabling agents to dynamically adapt and discover tool interfaces at runtime without restarting.
- Encrypted QR Mobile Handoff: Built end-to-end encrypted tunneling bridge pairing via dynamic QR code, allowing operators to monitor agent CLI terminal execution in real time from mobile devices. Packaged for macOS DMG, Windows MSI, Linux.

### DevisBat (swissquote-ai) — Commercial Swiss Trade Quoting & Fintech SaaS

**Next.js 15 / Supabase RLS / Gemini Vision / Swiss QR-Bill**

- Production Swiss B2B SaaS ($4,000 USD Contract): Architected and shipped commercial platform for Swiss client (live at devisbat.ch), converting technician voice notes and site photos into priced quotes and legally compliant Swiss QR-bill invoices across 12 trade categories, cutting quote turnaround from 4 hours to 90 seconds.
- 16,800+ SKU Matching Engine: Engineered catalogue matching engine against a 16.8k item database with automated Gemini Vision estimation fallback, Swiss canton wage tables, travel surcharges, and Supabase Row-Level Security.
- Fintech & Statutory Compliance: Integrated Swiss canton-specific labour wage tables, travel surcharges, material margin rules, Swiss VAT (8.1%), and automated QR-bill invoice generation with multi-tenant company isolation and trial lifecycle state machines.

### AstraQuote Lead Engine — Concurrency Pipeline & Verification Engine

**Python / Redis SETNX / DuckDuckGo / E.164 / Open Source**

- Atomic Concurrency Control: Architected multi-source ingestion pipeline with canonical SHA-256 fingerprinting and atomic Redis locks (SETNX), eliminating 100% of race conditions and duplicate API spend across 50,000+ operations.
- Deterministic Verification: Enforced zero-hallucination pipelines via automated E.164 phone formatting, Swiss canton validation, and Dead-Letter Queues (DLQ) with automated Slack alerting. Code audited and open-source on GitHub.
- Resilience & Rate-Limiting: Deployed token-bucket rate limiters with exponential backoff and jitter across multi-source discovery scrapers, maintaining 99.98% extraction reliability under aggressive target anti-bot throttling.

### Evolnex Relay — AES-256 Encrypted Omnichannel Edge Gateway

**Next.js Edge Runtime / Upstash Redis / WhatsApp Cloud API**

- Zero-Downtime Edge Routing: Built sub-80ms edge messaging gateway integrating WhatsApp Cloud API, Telegram Business API, and Gmail OAuth on Vercel Edge Runtime with record-level AES-256-GCM encryption and shadow-mode AI triage.
- Enterprise Security Architecture: Built record-level AES-256-GCM encryption with cloud-separated keys, webhook HMAC signature validation, intelligent shadow-mode AI triage, and fail-safe human takeover controls.
- Distributed Edge Streaming: Processed high-volume webhook bursts using Upstash Redis streaming queues on Vercel Edge Runtime, guaranteeing sub-80ms p95 delivery to avoid provider retry storms.

## PROFESSIONAL EXPERIENCE

### Founder & Lead Systems Architect — Evolnex Technologies

**2023 – Present | Remote**

- Direct technical architecture and engineering delivery for an agile 7-engineer software studio delivering bespoke enterprise AI platforms, high-throughput automation pipelines, and mission-critical web applications.
- Architected and shipped mission-critical web platforms, Progressive Web Apps (PWAs), and institutional proposals (including Aaranyak Conservation HRMS and West Bengal Primary LMS).
- Led development of proprietary SaaS products (DevisBat, Astra AI Core, Relay), managing full software development lifecycles from initial Figma UX design tokens to production Kubernetes/AWS cloud deployments.
- Established engineering standards across the team: engineered CI/CD pipelines, Dockerized runtime harnesses, deterministic LLM evaluation suites, and strict async/await concurrency patterns.

## KEY INSTITUTIONAL & CLIENT PLATFORMS

- Aaranyak Conservation HRMS: Architected 10-module enterprise HRMS & field appraisal platform for UNESCO World Heritage zones, featuring offline-first biometric authentication, GPS geofencing, and automated payroll pipelines.
- West Bengal Primary LMS Platform: Designed dual-portal learning management platform (11,000+ teachers, 100k+ students) with sub-100ms assessment evaluation engines and EduRev-compliant real-time data sync.
- Fountain Life HealthTech: Engineered high-conversion marketing platform for premier longevity membership brand (Longevity Clinic of the Year 2025) reaching high-net-worth members globally.
- Girl Effect (WAZZII Bot): Architected multi-lingual WhatsApp conversational bot and content moderation filters for adolescent reproductive health across East Africa, ensuring zero-hallucination statutory safety.

## CERTIFICATIONS & PROFESSIONAL CREDENTIALS

### Certificate in Automation and Agentic AI

**Udemy | 2026**

Advanced curriculum covering autonomous agentic execution loops, multi-agent orchestration, tool calling, and enterprise workflow design.

### Certificate in Full-Stack Software Engineering

**Udemy | 2026**

Comprehensive engineering credential covering Next.js 15, React 19, TypeScript, Python FastAPI, PostgreSQL RLS, REST/GraphQL APIs, and cloud deployments.

### Registered Micro Enterprise (MSME Udyam)

**Govt. of India | Verified**

UDYAM-UP-01-0202485 — Officially registered Indian digital engineering and software development enterprise.